GAS EXPORT COMPRESSOR VER 1

Operator Training Simulator · K-100 two-section export compressor driven by GT-100

Department of Chemical Engineering, Chulalongkorn University (teaching)

CAUSE AND EFFECT DIAGRAM

DOCUMENT NO. : KC-V1-IN-CE-001

DOCUMENT TITLE : CAUSE AND EFFECT DIAGRAM

P&ID : KC-PID-100 rev A

PFD : KC-PFD-100 rev A

SRS : KC-V1-IN-SRS-001

SHEETS : 1 (A3 landscape)

Teaching document generated from the simulator logic: bmsTrip(), tripConditions(), sisStep(), bmsCmd() (reset, ESDV-101 and XV-001 permissives), purgePermissives() and the GT start sequence in step(), and the alarm table ALM. Settings, effects and delays are checked against the model in the C&E matrix tab (section 5). Not for construction or operation of a real plant.

Rev.	Date	Revision status	Prepared	Checked	Approved	Remark
R0	12-Oct-26	Issued for teaching	Apinan Soottitantawat	—	—	

GAS EXPORT COMPRESSOR VER 1 · K-100 / GT-100
CAUSE and EFFECT DIAGRAM

Department of Chemical Engineering, Chulalongkorn University (teaching)

CLIENT :	Department of Chemical Engineering, Chulalongkorn University (teaching)	**REV**	**DATE**	**ORIGINATOR**	**CHECKER**	**APPROVER**
PROJECT :	Gas Export Compressor Ver 1 · OTS (K-100 / GT-100)	R0	12-Oct-26	Apinan Soottitantawat	—	—
DOCUMENT No. :	KC-V1-IN-CE-001					
DOCUMENT TITLE :	Cause and Effect Diagram					
P&ID / PFD :	KC-PID-100 rev A / KC-PFD-100 rev A					SHEET 1 of 1

CAUSES

EFFECTS

Item No.	Tag No.	Tag Service Description	Trip/Alarm Direction	Setting	Eng. Unit	Delay	Note No.	**XV-100** GT-100 fuel shut-off valve (fuel cut, GT coasts down)	**ESDV-101** Suction ESD valve, production to K-100	**ESDV-113** Export ESD valve, K-100 to the pipeline	**UV-103** Section 1 anti-surge valve: UIC-103 MAN 100 %	**UV-108** Section 2 anti-surge valve: UIC-108 MAN 100 %	**PIC-101** Suction pressure controller to MAN (cascade broken)	**SIC-100** Speed controller to MAN, output 0 %	**PV-102** PIC-102 trip feed-forward: output ≥ 80 % to flare	**BDV-101** Blowdown valve, suction side (V-101) to flare	**BDV-112** Blowdown valve, discharge side (E-102) to flare	**XV-001** Production inlet valve: wells shut in	**I-1** Compressor trip latch I-1 (SIS TRIPPED)	**1ST** First-out annunciation (SIS tab)	**START** GT-100 start command and sequence	**RESET** SIS reset command	**TRIP** Trip / status alarm in the DCS alarm list	**ALM** Process alarm in the DCS alarm list	**HORN** Audible alarm (horn)
							Trip Action	C	C	C	O	O	MAN	MAN	FF	O	O	C	TP	F	P	RB	A	A	A
001	**ESD-1**	Emergency shutdown push-button ESD-1 (SIS tab): compressor trip I-1	Pushed	—	—	0 s	1, 2, 3	X	X	X	X	X	X	X	X				X	X			X		X
002	**ESD-2**	Emergency shutdown push-button ESD-2: I-1 with blowdown and wells shut-in (I-2)	Pushed	—	—	0 s	2, 3, 6	X	X	X	X	X	X	X	X	X	X	X	X	X			X		X
003	**LSHH-101**	V-101 inlet scrubber liquid level high-high (true level, own taps)	High-High	85	%	3 s	2, 3, 4, 8	X	X	X	X	X	X	X	X				X	X			X		X
004	**LSHH-107**	V-102 interstage scrubber liquid level high-high (true level, own taps)	High-High	85	%	3 s	2, 3, 4, 8	X	X	X	X	X	X	X	X				X	X			X		X
005	**TSHH-104**	Section 1 discharge temperature high-high on TI-104 (seal ceiling)	High-High	150	°C	3 s	2, 3, 4	X	X	X	X	X	X	X	X				X	X			X		X
006	**TSHH-109**	Section 2 discharge temperature high-high on TI-109 (seal ceiling)	High-High	150	°C	3 s	2, 3, 4	X	X	X	X	X	X	X	X				X	X			X		X
007	**VSHH-105**	Section 1 shaft vibration high-high on VI-105	High-High	90	µm	3 s	2, 3, 4	X	X	X	X	X	X	X	X				X	X			X		X
008	**VSHH-110**	Section 2 shaft vibration high-high on VI-110	High-High	90	µm	3 s	2, 3, 4	X	X	X	X	X	X	X	X				X	X			X		X
009	**PSLL-150**	Lube-oil header pressure low-low on PI-150 while the train turns	Low-Low	1.2	barg	2 s	2, 3, 4, 5	X	X	X	X	X	X	X	X				X	X			X		X
010	**SSHH-100**	Overspeed on the governor speed signal (SIC-100 measurement)	High-High	110	%	0.5 s	2, 3, 4, 9	X	X	X	X	X	X	X	X				X	X			X		X
011	**PSHH-112**	Discharge header pressure high-high on PI-112 (after E-102)	High-High	132	barg	2 s	2, 3, 4	X	X	X	X	X	X	X	X				X	X			X		X
012	**PSLL-101**	Suction pressure low-low on the PIC-101 measurement (V-101), armed in RUN	Low-Low	20	barg	5 s	2, 3, 4, 5, 9	X	X	X	X	X	X	X	X				X	X			X		X
013	**GT-100**	GT-100 trip / flame-out (GT unit control panel)	Trip	—	—	0 s	2, 3, 4	X	X	X	X	X	X	X	X				X	X			X		X
014	**PSHH-001**	Production header pressure high-high on PT-001: XV-001 shuts in the wells (no I-1)	High-High	34.5	barg	3 s	4, 7, 9											X					X		X
015	**PIC-102**	Production header pressure high: gas flaring	High	32.5	barg		10, 12																	X	X
016	**FI-109**	Total gas to flare	High	5	t/h		10																	X	X
017	**PIC-101**	Suction pressure low (V-101)	Low	28.5	barg		10, 11																	X	X
018	**PIC-101**	Suction pressure high (V-101)	High	31.5	barg		10, 11																	X	X
019	**LIC-101**	V-101 liquid level low	Low	20	%		10																	X	X
020	**LIC-101**	V-101 liquid level high (pre-alarm of LSHH-101)	High	65	%		10																	X	X
021	**LIC-107**	V-102 liquid level low	Low	20	%		10, 11																	X	X
022	**LIC-107**	V-102 liquid level high (pre-alarm of LSHH-107)	High	65	%		10, 11																	X	X
023	**UIC-103**	Section 1 surge margin low	Low	7	%		10, 11																	X	X
024	**UIC-108**	Section 2 surge margin low	Low	7	%		10, 11																	X	X
025	**TI-104**	Section 1 discharge temperature high (pre-alarm of TSHH-104)	High	125	°C		10, 11																	X	X
026	**TI-109**	Section 2 discharge temperature high (pre-alarm of TSHH-109)	High	130	°C		10, 11																	X	X
027	**TIC-106**	Intercooler E-101 gas outlet temperature low	Low	30	°C		10, 11																	X	X
028	**TIC-106**	Intercooler E-101 gas outlet temperature high	High	48	°C		10, 11																	X	X
029	**TIC-111**	Export gas temperature low (E-102 outlet)	Low	30	°C		10, 11																	X	X
030	**TIC-111**	Export gas temperature high (E-102 outlet)	High	55	°C		10, 11																	X	X
031	**PI-112**	Discharge pressure high (pre-alarm of PSHH-112)	High	126	barg		10, 11																	X	X
032	**VI-105**	Section 1 vibration high (pre-alarm of VSHH-105)	High	60	µm		10, 11																	X	X
033	**VI-110**	Section 2 vibration high (pre-alarm of VSHH-110)	High	60	µm		10, 11																	X	X
034	**SIC-100**	Compressor speed low	Low	72	%		10, 11																	X	X
035	**SIC-100**	Compressor speed high (pre-alarm of SSHH-100)	High	104	%		10, 11																	X	X
036	**PI-150**	Lube-oil pressure low (pre-alarm of PSLL-150)	Low	1.8	barg		10, 13																	X	X
037	**TI-151**	Bearing temperature high (section 2 end)	High	95	°C		10, 11																	X	X
038	**TI-100**	GT exhaust temperature high	High	650	°C		10, 11																	X	X
039	**AI-115**	Export gas molar mass high (rich gas)	High	23.5	kg/kmol		10																	X	X
040	**FI-114**	Export flow low	Low	150	MMSCFD		10, 11																	X	X
041	**I-1**	GT start permissive: SIS reset (READY)	Not reset	—	—		14														P				
042	**PI-150**	GT start permissive: Lube oil PI-150 ≥ 2.0 barg	Low	2	barg		14														P				
043	**P-160**	GT start permissive: Cooling medium pump P-160 running	Stopped	—	—		14														P				
044	**ESDV-101**	GT start permissive: ESDV-101 open, suction PI-101 ≥ 20 barg	Low	20	barg		14														P				
045	**ZI-103/108**	GT start permissive: UV-103 and UV-108 ≥ 90 % open (full recycle)	Low	90	%		14														P				
046	**LIC-101/107**	GT start permissive: Scrubber levels below 60 %	High	60	%		14														P				
047	**BDV-101/112**	GT start permissive: BDVs closed	Open	—	—		14														P				
048	**GT-100**	Start accepted: XV-100 opens, GT-100 cranks and purges (CRANK)	Start	—	—	30 s	14	O													S				
049	**GT-100**	Ignition, flame proven, then acceleration (IGN → ACCEL)	Start	—	—	10 s	14														S				
050	**SIC-100**	Minimum governor speed: unit in RUN, SIC-100 to AUTO at 70 %	High	69.5	%		14							AUT							S				
051	**SIS RESET**	Reset refused while a trip condition holds (PSLL-101 excepted) or the train turns	Turning	5	%		15															RB			
052	**ESDV-101**	Opening permissive: ΔP across ESDV-101 (pressurise through HV-101 first)	High	3	bar		15		P																
053	**PI-001**	XV-001 reopen (wells back on line) blocked while the header stays high	High	33.5	barg		7											RB							
054																									
055																									

Notes:

1. Action codes: C close; O open; MAN controller forced to manual; FF feed-forward; TP tripped and latched (manual reset on the SIS tab); F first-out; P permissive (the start or the opening is refused while the condition holds); RB reset or reopen blocked; S sequence step; AUT controller to AUTO; A activate. X marks the column's action.
2. I-1 compressor trip (bmsTrip): XV-100 shuts the fuel and GT-100 coasts down, ESDV-101 and ESDV-113 shut (4 s stroke in the model), UIC-103/108 go to MAN 100 % (UV-103/108 open), PIC-101 and SIC-100 go to MAN with the SIC-100 output at 0 %. The first cause is latched as the first-out until reset. The loop then settles out at ≈ 69.7 barg (model, ESD-1).
3. PV-102 feed-forward: bmsTrip raises the PIC-102 output to at least 80 % and re-initialises the controller only when PIC-102 is in AUTO; PIC-102 then keeps controlling the production header from that output (production to flare).
4. Delays are counted in the logic solver at the 0.25 s scan; a trip acts at the first scan after its delay (3.25, 2.25 and 0.75 s in the model). A trip condition acts only while the SIS is in RUN or START. The GT flame-out also trips I-1 directly from the GT logic, with no delay.
5. PSLL-101 is armed only in RUN and above 60 % speed, so a start-up is not tripped on low suction, and it does not block a reset. PSLL-150 acts only while the train turns (> 5 % speed).
6. ESD-2 = I-2: I-1 plus BDV-101 and BDV-112 open to flare and XV-001 shuts the wells. It acts even after I-1 has latched (the first-out stays the first cause). Model: ESDVs shut at 4 s, loop below 6.9 barg after 6.1 min.
7. PSHH-001 shuts XV-001 only: K-100 keeps running until the falling suction reaches PSLL-101 or the operator acts. XV-001 reopens on command only below 33.5 barg.
8. LSHH-101 and LSHH-107 read the true level on their own taps, not the LIC-101 / LIC-107 transmitters, so they are independent of the level controllers.
9. Shared measurements: SSHH-100 reads the governor speed signal of SIC-100, PSLL-101 reads PT-101 of PIC-101, PSHH-001 reads PT-001 of PIC-102 (see the SRS and LOPA tabs).
10. Alarms have no delay; they clear with a 1 % deadband. Every new alarm sounds the horn until silenced and enters the alarm list for acknowledgement.
11. Suppressed unless the unit is in RUN above 60 % speed.
12. Suppressed unless the wells produce (XV-001 open, production above 5 kg/s).
13. Suppressed unless the train turns or a lube-oil pump runs.
14. GT-100 start: with every permissive met the start command opens XV-100, the GT cranks and purges for 30 s, ignites for 10 s, accelerates and enters RUN at 69.5 % speed (SIC-100 AUTO, SP 70 %). The start is refused while the GT flame-out fault is still active.
15. Reset (one command on the SIS tab) is refused while any trip condition except PSLL-101 holds or the train turns above 5 %. The ESDVs then stay shut until opened one by one: ESDV-101 only with ΔP ≤ 3 bar (pressurise through HV-101), ESDV-113 on command.
16. Not SIS actions: P-150B auto-starts at 1.8 barg (BPCS); P-160 stopped, ESDV-113 shut while running, surge, GT at its power limit and stonewall are status alarms in the DCS alarm list without a setting, so they are not listed as causes.

Generated from the K-100 simulator logic (SIS trips, first-out, blowdown, wells shut-in, GT start permissives and sequence, reset rules) and the alarm table; settings, effects and delays agree with the model (C&E matrix tab, section 5). Teaching document — not for construction.